$[1, \infty)$

Use the integral test to determine if $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{8}{3}}}$ converges or diverges.

Integral test → If a_n is positive, continuous, and decreasing, then

$\sum_{n=b}^{\infty} a_n$ diverges if $\int_b^{\infty} f(n)\,dn$ diverges,

and if

$\sum_{n=b}^{\infty} a_n$ converges if $\int_b^{\infty} f(n)\,dn$ converges.

$$\sum_{n=1}^{\infty} \frac{1}{n^{8/3}}$$

Let $a_n = f(x)$

$$\int_1^{\infty} \frac{1}{x^{8/3}}\,dx$$

$$\lim_{b\to\infty} \int_1^{b} x^{-8/3}\,dx$$

$$\lim_{b\to\infty} \left[\frac{3x^{-5/3}}{-5}\right]_1^b$$

$$\lim_{b\to\infty} \left[\frac{3b^{-5/3}}{-5} + \frac{3(1)^{-5/3}}{5}\right]$$

$$\lim_{b\to\infty} \left[\frac{3}{-5}\left(\frac{1}{b^{5/3}}\right) + \frac{3}{5}\right]$$

0

$\frac{1}{\text{very large \#}} \Rightarrow 0$

$$(0) + \frac{3}{5}$$

$$\boxed{= \frac{3}{5}}$$

* If the integral converges to some value K, the infinite series also converges, but not <u>necessarily to K</u>.

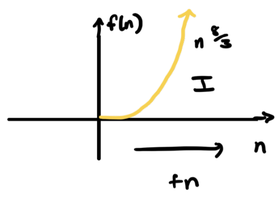
(Assuming that the integral test is applied legally).

The integral test legally proves that $\sum_{n=1}^{\infty} \frac{1}{n^{8/3}}$ converges.

Prove:

a_n is positive on $[1, \infty)$

$$\frac{1}{n^{8/3}} > 0$$



$n^{8/3} > 0$ for $n \geq 1$.

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Prove:

a_n is continuous on $[1, \infty)$

$\frac{1}{n^{8/3}}$ $\quad n^{8/3} = 0$ when $n = 0$ the function $f(n)$, or a_n is discontinuous.

when $n=0$, then $f(n)$ is discontinuous. $\Rightarrow$:(

$n=0$ is not on $[1, \infty)$ $\Rightarrow$:)

Integral test can still be applied

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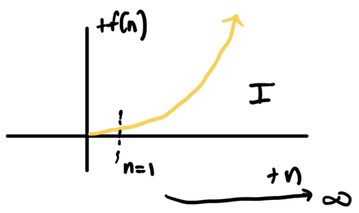
Prove:

~~a_n~~ $f(n)$ is decreasing on $[1, \infty)$.

$$f(n) = \frac{1}{n^{8/3}}$$

$$f'(n) = \left[n^{-8/3}\right]'$$

$$= \left[\frac{-8}{3} n^{-11/3}\right] = \frac{-8}{3n^{11/3}} < 0 \text{ for } n \geq 1$$



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